Bézout's Identity

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1 Introduction

This handout contains problems for the Number Theory course taught by the Gauss Group. The problems are ordered from low difficulty to high difficulty. Some of these problems will be gone over during class. There are definitely a lot of problems, so feel free to pick and choose whatever looks interesting to you! If you need any clarification or spot an error, please email us.

2 Problems

Theorem: For $a, b \in \mathbb{N}$ and $x, y \in \mathbb{Z}$, there exist integers (x, y) such that

$$ax + by = \gcd(a, b)$$

If (a, b) is given, then (x, y) can be found using the Reverse Euclidean Algorithm.

1. Find the Bézout's Identity for $(a, b) = (4, 5)$.
2. Find the Bézout's Identity for $(a, b) = (9, 12)$.
3. Find the Bézout's Identity for $(a, b) = (15, 12)$.
4. Find the Bézout's Identity for $(a, b) = (27, 45)$.
5. Find the Bézout's Identity for $(a, b) = (102, 38)$.
6. Find the Bézout's Identity for $(a, b) = (34, 19)$.
7. Find the Bézout's Identity for $(a, b) = (2014, 4021)$.
8. Find the Bézout's Identity for $(a, b) = (237, 13)$.
9. Find the Bézout's Identity for $(a, b) = (177741, 149553)$.
10. Find the Bézout's Identity for $(a, b) = (6 \cdot 27, 6 \cdot 81)$.
11. Show that if a, b, c are integers such that $\gcd(a, c) = 1$, $\gcd(a, c) = 1$, and $\gcd(b, c) = 1$, then $\gcd(ab, c) = 1$.
12. If $\gcd(a, b) = d$, then find a formula for $\gcd(ka, kb)$ for some integer k.
13. Find (x, y) such that $36x + 20y = 8$.

14. Find (x, y) such that $45x + 12y = 6$.
15. Find (x, y) such that $24x + 18y = 12$.
16. Find (x, y) such that $27x + 16y = 4$.
17. Let $ax + by = 12$. How many possible values of $\gcd(a, b)$ are there?
18. Let $ax + by = 94$. How many the possible values of $\gcd(a, b)$ are there?
19. Let $ax + by = 292$. How many the possible values of $\gcd(a, b)$ are there?
20. Let $ax + 3by = 63$. How many the possible values of $\gcd(a, 3b)$ are there?
21. Prove with Bézout's Identity that if a and b are co-prime integers, then if $a|c$ and $b|c$, then $ab|c$.
22. Prove with Bézout's Identity that if a and b are co-prime integers, then if $a|bc$ then $a|c$. Then as a review, try proving it with the Fundamental Theorem of Arithmetic.
23. Let a and b be two natural numbers. Show with Bézout's Identity that if $\gcd(a, b) = 1$, then $\gcd(a, b^2) = 1$.
24. Let a and b be two natural numbers. Show with Bézout's Identity that if $\gcd(a, b) = 1$, then $\gcd(a^2, b^2) = 1$.
25. **(1959 IMO)** Show that
$$\frac{21N + 4}{14N + 3}$$
cannot be reduced further for every integer N.
26. **(1987 AHSME)** ABC is a triangle: $A = (0, 0), B = (36, 15)$ and both the coordinates of C are integers. What is the minimum area $\triangle ABC$ can have?
27. **(2006 AMC 12A)** Two farmers agree that pigs are worth 300 dollars and that goats are worth 210 dollars. When one farmer owes the other money, he pays the debt in pigs or goats, with "change" received in the form of goats or pigs as necessary. (For example, a 390 dollar debt could be paid with two pigs, with one goat received in change.) What is the amount of the smallest positive debt that can be resolved in this way?